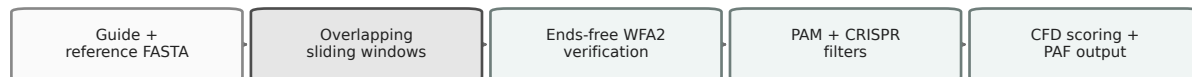
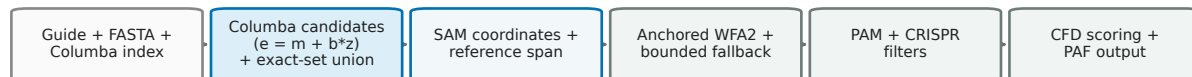


A Integration workflow

Original CRISPRapido

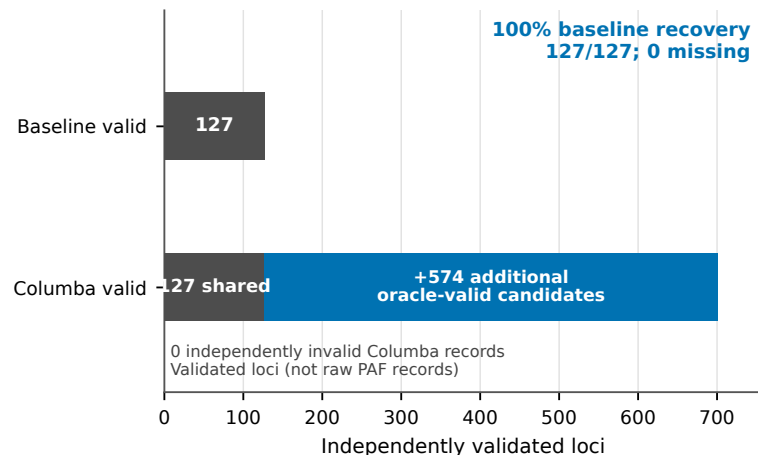


Columba-enabled CRISPRapido

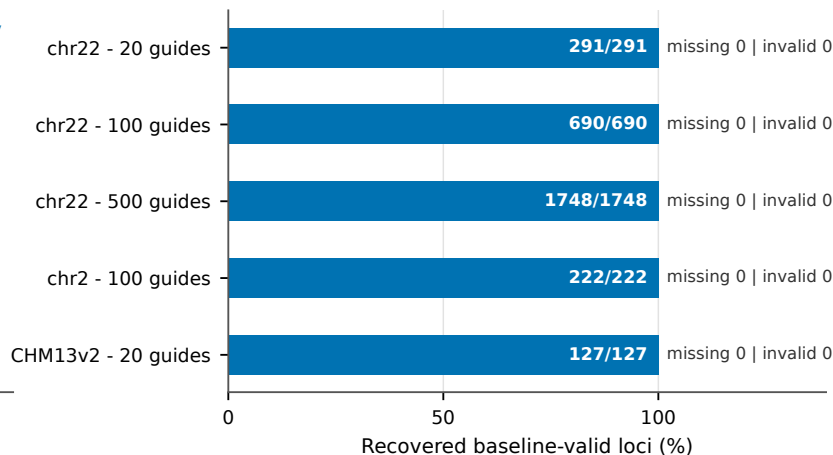


Columba replaces candidate generation; WFA2 verification, PAM/filtering, CFD scoring, and PAF reporting remain downstream.

B Whole-genome correctness



C Correctness across benchmark scales



Controlled synthetic: 6/6 candidate-matched configurations produced byte-identical PAF (separate raw-output test).